Year 13 Further Maths - Polar coordinates

Similar questions - Conversions, intersections and maximum distance

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve has Cartesian equation $x^3 + y^3 = 3xy$. The portion C with $x \geqslant 0$, $y \geqslant 0$ has polar equation $r = \mathrm{f}(\theta)$ for $0 \leqslant \theta \leqslant \frac{\pi}{2}$.

(a) Find $\mathrm{f}(\theta)$. **[2]**

(b) Show that $\mathrm{f}\left(\frac{\pi}{2} - \theta\right) = \mathrm{f}(\theta)$ and hence state the line of symmetry of C. **[3]**

(c) Find the value of r when $\theta = \frac{\pi}{4}$. **[1]**

(d) By finding values of θ for which $r = 0$, show that C has a loop. **[2]**

2 Two polar curves are $C_1 : r = 3\theta$ and $C_2 : r = \theta + 2$ for $0 \leqslant \theta \leqslant 2\pi$.

(a) C_1 meets the initial line at the pole and at a point A. Write down the polar coordinates of A. **[2]**

(b) Determine the polar coordinates of the point where C_1 and C_2 intersect. **[2]**

(c) Sketch C_2 on a copy of the sketch of C_1. **[1]**

3 The curve C has polar equation $r = 3\sin(2\theta)\sinh\left(\frac{1}{2}\theta\right)$ for $0 \leqslant \theta \leqslant \frac{1}{2}\pi$.

(a) Explain how you can tell that C is a single loop in the first quadrant passing through the pole. **[3]**

(b) The point on C furthest from the pole is A where $\theta = \phi$. Show that ϕ satisfies $\phi = \ln\left(\dfrac{4 - \tan 2\phi}{4 + \tan 2\phi}\right)$. **[4]**

(c) You are given that the relevant solution is $\phi = 1.0284$ (4 d.p.). Find the distance from A to the pole to 3 significant figures. **[1]**

Question 1

(a) $r^3(\cos^3\theta+\sin^3\theta)=3r^2\sin\theta\cos\theta$, so $r=\dfrac{3\sin\theta\cos\theta}{\cos^3\theta+\sin^3\theta}$

(b) Swapping $\sin\theta$ and $\cos\theta$ leaves f unchanged, so $\mathrm{f}\left(\frac{\pi}{2}-\theta\right)=\mathrm{f}(\theta)$ and C is symmetric about the line $y=x$ ($\theta=\frac{\pi}{4}$)

(c) $r=\dfrac{3\cdot\frac{1}{2}}{2\left(\frac{\sqrt{2}}{2}\right)^3}=\dfrac{3}{\sqrt{2}}=2.12$

(d) $r=0$ when $\sin\theta=0$ or $\cos\theta=0$, i.e. $\theta=0$ and $\theta=\frac{\pi}{2}$, with $r>0$ in between. So the curve starts and ends at the pole and forms a loop

Question 2

(a) C_1 meets the initial line when $\theta=0$ (pole) or $\theta=2\pi$: $A=(6\pi,2\pi)$

(b) $3\theta=\theta+2$ gives $\theta=1$, $r=3$. The point is $(3,1)$

(c) C_2 is a spiral starting at $(2,0)$ with r increasing steadily to $2\pi+2$ at $\theta=2\pi$; it starts outside C_1 (which starts at the pole), crosses it at $(3,1)$, then lies inside C_1

Question 3

(a) $r\geqslant 0$ for $0\leqslant\theta\leqslant\frac{\pi}{2}$ since $\sin 2\theta\geqslant 0$ and $\sinh\frac{\theta}{2}\geqslant 0$. $r=0$ at $\theta=0$ and $\theta=\frac{\pi}{2}$ and nowhere between, so the curve starts and ends at the pole and has positive r in $(0,\frac{\pi}{2})$, so lies in the first quadrant as one loop

(b) $\dfrac{\mathrm{d}r}{\mathrm{d}\theta}=6\cos 2\theta\sinh\frac{\theta}{2}+\frac{3}{2}\sin 2\theta\cosh\frac{\theta}{2}=0$. Dividing by $\cosh\frac{\theta}{2}$: $6\cos 2\theta\tanh\frac{\theta}{2}+\frac{3}{2}\sin 2\theta=0$, so $\tanh\frac{\theta}{2}=-\dfrac{\tan 2\theta}{4}$.
Using $\tanh^{-1}u=\frac{1}{2}\ln\dfrac{1+u}{1-u}$: $\dfrac{\theta}{2}=\frac{1}{2}\ln\dfrac{4-\tan 2\theta}{4+\tan 2\theta}$, so $\phi=\ln\dfrac{4-\tan 2\phi}{4+\tan 2\phi}$

(c) $r=3\sin(2\phi)\sinh\frac{\phi}{2}=1.42$